

Root Locus

1. It plots the system's Dynamic Characteristics.
2. It explains the plot of systems roots (Poles & zeros) on s-plane
3. It explains the system's response and region of stability on s-plane

Rules of Root Locus

Root locus plot emerges from the poles and gets enclosed to the zeros or asymptotes.

Root locus must be symmetric with respect to the real axis.

All steps are not compulsory, but based on requirements we need to calculate steps.

1. Identify the roots (poles and zeros) of the system. Total loci = $\text{Max}(P, Z)$

2. No. of asymptotes $X = \text{Total Poles} - \text{total zeros}$

3. Angle of asymptotes $= \frac{(2n+1)}{P-Z} 180^\circ$; $n = 0, 1, 2, \dots$

4. Centroid of asymptotes, $\sigma_c = \frac{\sum \text{real part of Poles} - \sum \text{real part of zeros}}{P-Z}$

5. Find Break Away Points

i) Find ch. eqn $[1 + G(s)H(s) = 0]$

ii) Compute $k = \text{polynomial}$.

iii) Compute $\frac{dk}{ds} = 0$ and find $s = \text{B.A.P}$

6. Find angle of departure, $\theta_d = 180^\circ - [\sum \theta_p - \sum \theta_z]$

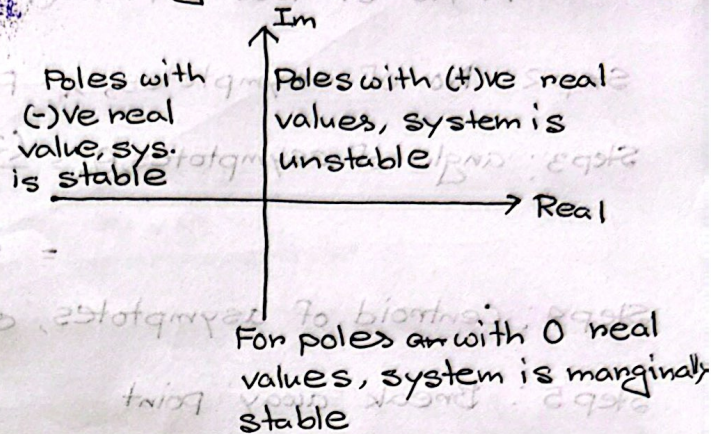
7. Intersection of imaginary axis.

i) Find ch. eqn.

ii) Construct Routh matrix

iii) Find k for marginal stable system. $\{1^{\text{st}} \text{ clmn Routh } M\}$

iv) Place k in auxiliary eqn $\{2^{\text{nd}} \text{ order eqn in Routh } M\}$



1. Obtain Root-locus plot for a unity feedback system with system Gain!

$$G(s) = \frac{k}{s(s+2)}$$

Step 1: Obtain loci - $P_1 = 0$, $P_2 = -2$ and no zeros

$$\therefore \text{No. of loci} = \text{Max}(P, Z) = 2$$

Step 2: No. of asymptotes, $X = P - Z = 2$

Step 3: angle of asymptotes, $\theta_n = \frac{(2n+1)}{P-Z} 180^\circ$, $n=0, 1$

$$= 90^\circ, 270^\circ$$

Step 4: Centroid of asymptotes, $\sigma_c = \frac{\sum \text{real } P - \sum \text{real } Z}{P-Z} = \frac{-2-0}{2-0} = -1$

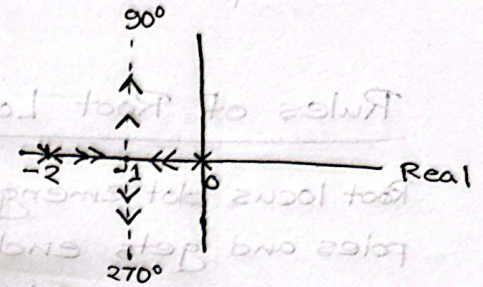
Step 5: Break away point

$$\text{Ch. eqn } 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{k}{s(s+2)} = 0$$

$$\Rightarrow k = -s(s+2) = -s^2 - 2s$$

$$\text{So, } \frac{dk}{ds} = 0 \Rightarrow -2s - 2 = 0 \Rightarrow s = -1$$



$$2. G(s)H(s) = \frac{k(s+2)}{s(s+1)(s+3)}$$

Poles = 0, -1, -3

Zeros = -2

No. of loci = 3

No. of asymptotes, $X = P - Z = 2$

angle of " , $\theta_n = \frac{(2n+1)}{P-Z} 180^\circ = 90^\circ, 270^\circ$

centroid of " , $\sigma_c = \frac{-3-1+2}{3-1} = -1$

Break away point

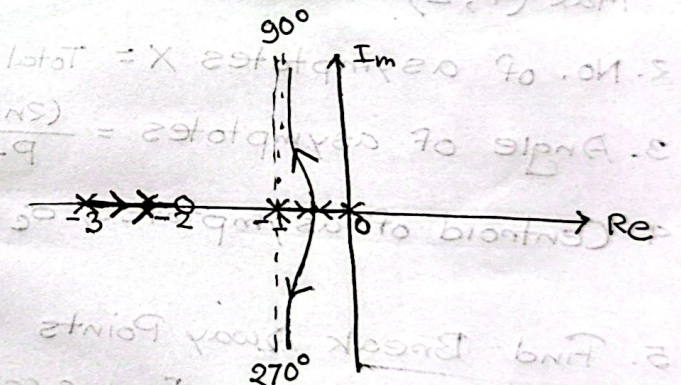
$$\text{ch. eqn } 1 + \frac{k(s+2)}{s(s+1)(s+3)} = 0$$

$$\Rightarrow k = \frac{-s^3 + 4s^2 + 3s}{s+2}$$

$$\frac{dk}{ds} = 0$$

$$\Rightarrow \frac{(s+2)(3s^2 + 8s + 3) - (s^3 + 4s^2 + 3s)}{(s+2)^2} = 0 \Rightarrow s^3 + 5s^2 + 8s + 3 = 0$$

$$\Rightarrow s = -0.53, -2.23 - j0.79, -2.23 + j0.79$$

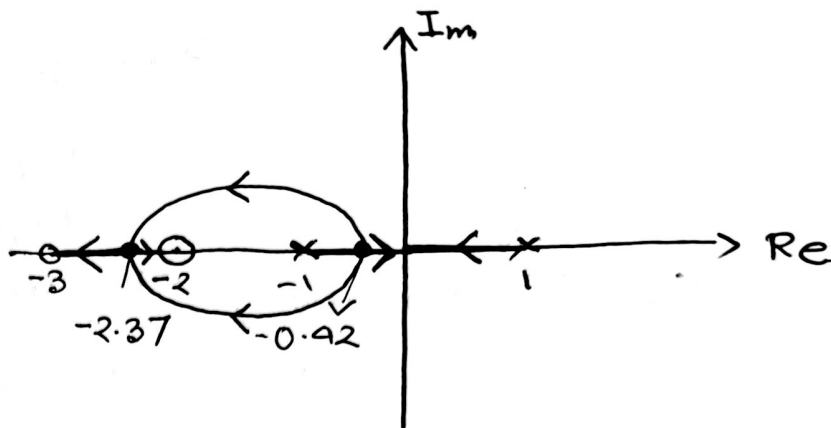


$$3. \quad G(s) = \frac{k(s+2)(s+3)}{(s+1)(s-1)}$$

$$1. \quad z_1 = -2, \quad z_2 = -3;$$

$$p_1 = -1, \quad p_2 = 1$$

$$\text{Total loci} = \max(P, Z) \\ = 2$$



3. BAP

$$\text{ch. eqn } 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{k(s+2)(s+3)}{(s+1)(s-1)} \times 1 = 0$$

$$\Rightarrow s^2 - 1 + k(s^2 + 5s + 6) = 0$$

$$\Rightarrow k = -\frac{s^2 - 1}{s^2 + 5s + 6}$$

$$\frac{d(A/B)}{ds} = \frac{B \frac{dA}{ds} - A \frac{dB}{ds}}{B^2}$$

$$\frac{dk}{ds} = 0 \Rightarrow \frac{(s^2 + 5s + 6)(2s) - (s^2 - 1)(2s + 5)}{(s^2 + 5s + 6)^2} = 0$$

$$\Rightarrow 2s^3 + 10s^2 + 12s - 2s^3 - 5s^2 + 2s + 5 = 0$$

$$\Rightarrow 5s^2 + 14s + 5 = 0$$

$$\Rightarrow s = \frac{-14 \pm \sqrt{14^2 - 4ac}}{2 \times 5}$$

$$= -0.42, -2.37$$

$$4. \quad G(s) = \frac{k}{s(s^2 + 6s + 10)}$$

Poles: $0, -3 + j, -3 - j$

$$s^2 + 6s + 10 = 0$$

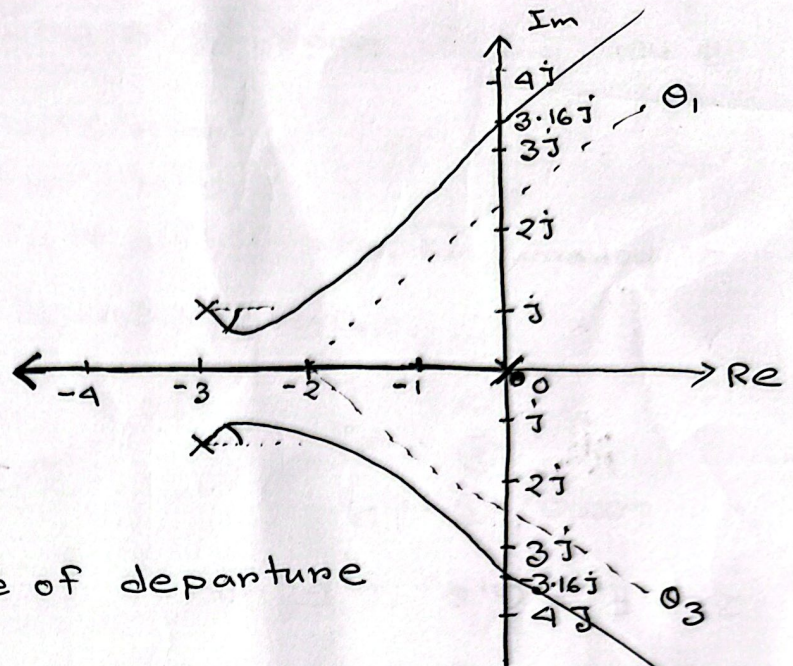
$$\Rightarrow s = \frac{-6 \pm \sqrt{36 - 40}}{2}$$

$$= -3 \pm j$$

Zeros: 0

$\therefore$ No. of loci = 3.

Complex pole, so angle of departure required.



No. of asymptotes, $X = \text{total no. of } P - \text{total no. of } Z$
 $= 3 - 0$

angle of " = $\frac{2n+1}{P-Z} \times 180^\circ$; $n=0,1,2$
 $= 60^\circ, 180^\circ, 300^\circ$

Centroid of " = $\frac{0 - 3 - 3 - 0}{3 - 0} = -2$

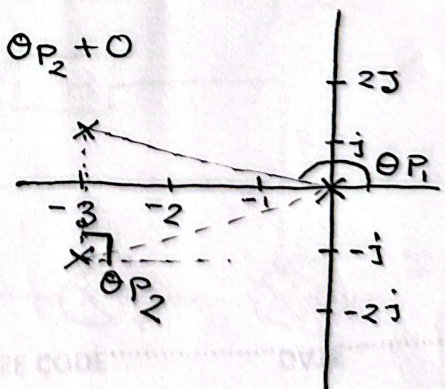
$$\theta_{d1} = 180^\circ - [\angle \theta_{P1} - \angle \theta_{Z1}] = 180^\circ - \theta_{P1} - \theta_{P2} + 0$$

$$= (180^\circ - \tan^{-1} \frac{1}{3}) - 90^\circ + 180^\circ$$

$$= (180^\circ - 18.435^\circ) - 90^\circ + 180^\circ$$

$$= -71.565^\circ$$

$$\theta_{d2} = 71.565^\circ$$



Now, $1 + G(s)H(s) = 0$

$$\Rightarrow 1 + \frac{k}{s^3 + 6s^2 + 10s} = 0$$

$$\Rightarrow s^3 + 6s^2 + 10s + k = 0$$

s^3	1	10
s^2	6	k
s^1	$\frac{60-k}{6}$	0
s^0	k	

$$\frac{60-k}{6} \geq 0$$

$$\Rightarrow k = 60 \quad [k=60, \text{ system marginally stable}]$$

$$6s^2 + k = 0$$

$$\Rightarrow 6s^2 + 60 = 0$$

$$\Rightarrow s^2 = -10$$

$$\Rightarrow s = \pm j 3.16$$